HOLC Red-lining and Environmental Justice Issues in LA

Jacob Sloan

2026-10-26

```
library(gt)
library(tidyverse)
library(sf)
library(here)
library(tmap)
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/home/jacobsloan/EDS223-hw-2/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.g
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/home/jacobsloan/EDS223-hw-2/data/mapping-inequality/mapping-inequality-
los-angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```
Reading layer `mapping-inequality-new-haven' from data source
  `/home/jacobsloan/EDS223-hw-2/data/mapping-inequality/mapping-inequality-
new-haven.json'
```

```
  using driver `GeoJSON'
Simple feature collection with 49 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -72.99321 ymin: 41.23109 xmax: -72.85227 ymax: 41.37502
Geodetic CRS:  WGS 84
```

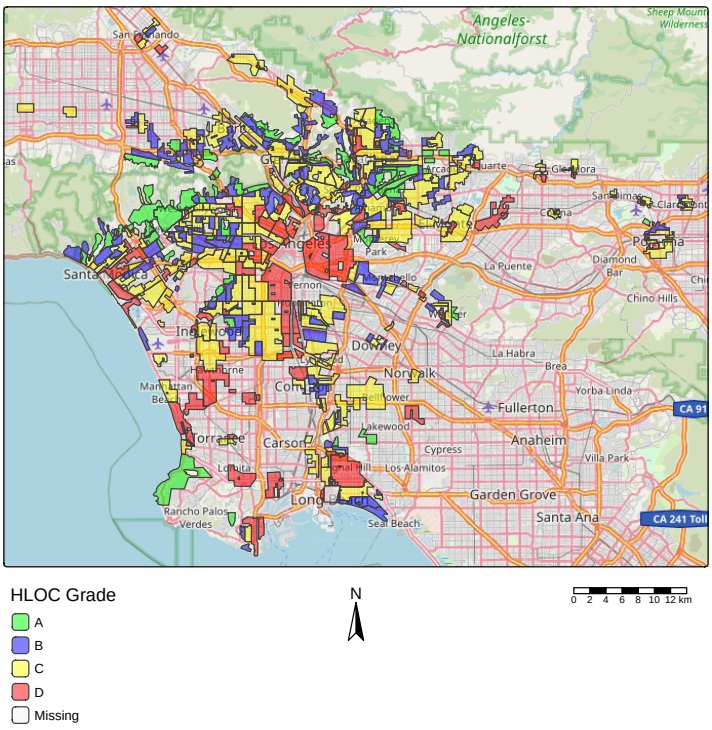
Part 1: Legacy of redlining in current environmental (in)justice

Create a map of historical redlined neighborhoods

```
# map of LA redlining

tm_basemap('OpenStreetMap.DE') +
tm_shape(mapineqlos) +
    tm_polygons(
        fill = 'grade',
        fill_alpha = 0.5,
        fill.scale = tm_scale_categorical(values = c('green','blue','yellow','red')),
        fill.legend = tm_legend('HLOC Grade', frame = FALSE,
            position = tm_pos_out('center','bottom','left'))
    )+
    tm_title(text = 'HOLC Ranking of Neighborhoods in LA') +
    tm_compass(type = 'arrow', position = tm_pos_out('center','bottom','center'))+
    tm_scalebar(position = tm_pos_out('center','bottom','right'))
```

HOLC Ranking of Neighborhoods in LA



Create table summarizing proportion of census block in each HOCL Grade

```
laejscreen <- ejscreen |>
    filter(CNTY_NAME == 'Los Angeles County')

# Check CRS's before spatially joining

st_crs(mapineqlos) == st_crs(laejscreen)
```

```
[1] FALSE
```

```
# convert CRS's and then spatial join
mapineqlos<- st_transform(mapineqlos, crs = st_crs(laejscreen))
ejxred <- st_join(laejscreen, mapineqlos, join = st_intersects)

# finding percent of blocks that are x Grade HOLC
```

Proportion of Census Blocks in each HOCL Grade

grade	proportion
A	0.0499555
B	0.1378505
C	0.3402314
D	0.1497552
NA	0.3222074

```
gradeprop <- ejxred |>
    st_drop_geometry() |>
    select('grade') |>
    count(grade)
gradeprop$n <- gradeprop$n/nrow(ejxred)
gradeprop |>
    rename(proportion= n) |>
    gt() |>
    tab_header('Proportion of Census Blocks in each HOCL Grade')
```

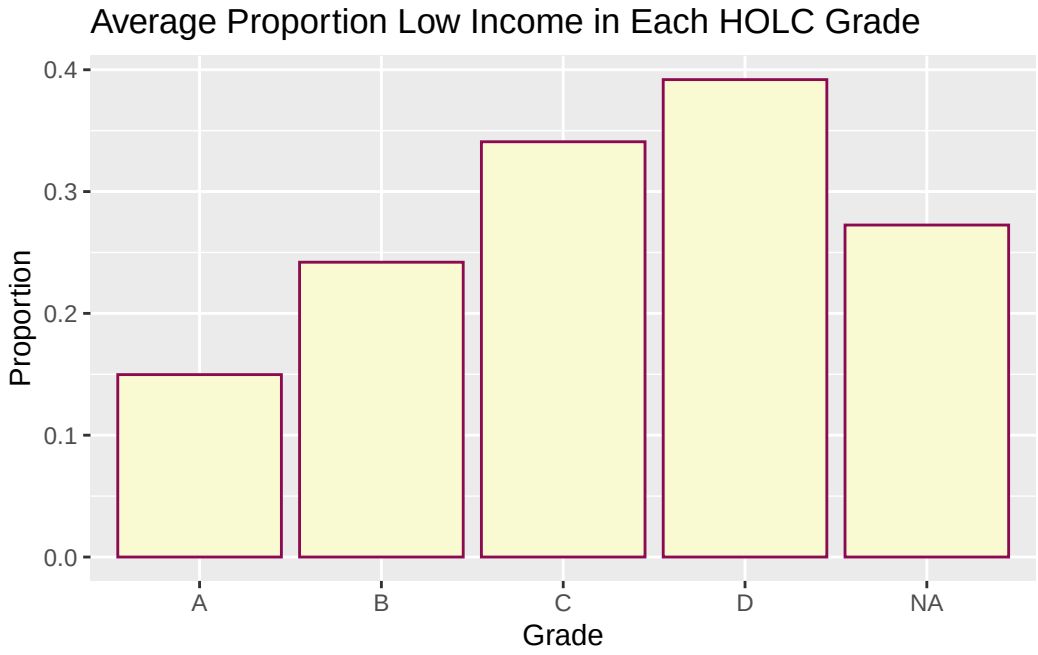
Create three visualizations summarizing current conditions

```
# Summarizing mean PM25 concentration, percent of people low income,
# and percent people with low life expectancy by HOCL Grade.

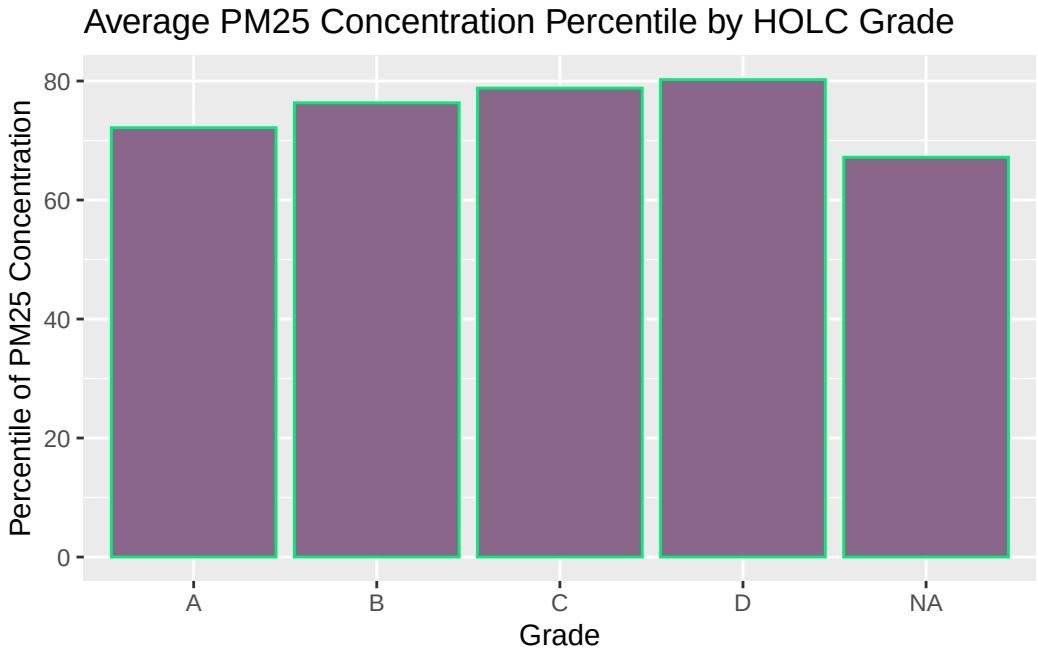
sumbygrade <- ejxred |>
    group_by(grade) |>
    summarise(meanlowinc = mean(LOWINCPCT, na.rm =TRUE),
        meanP_PM25 = mean(P_PM25, na.rm =TRUE),
        meanlowlifexpect = mean(P_LIFEEXPPCT, na.rm =TRUE))

# Plotting each EJSCREEN variable by HOCL grade

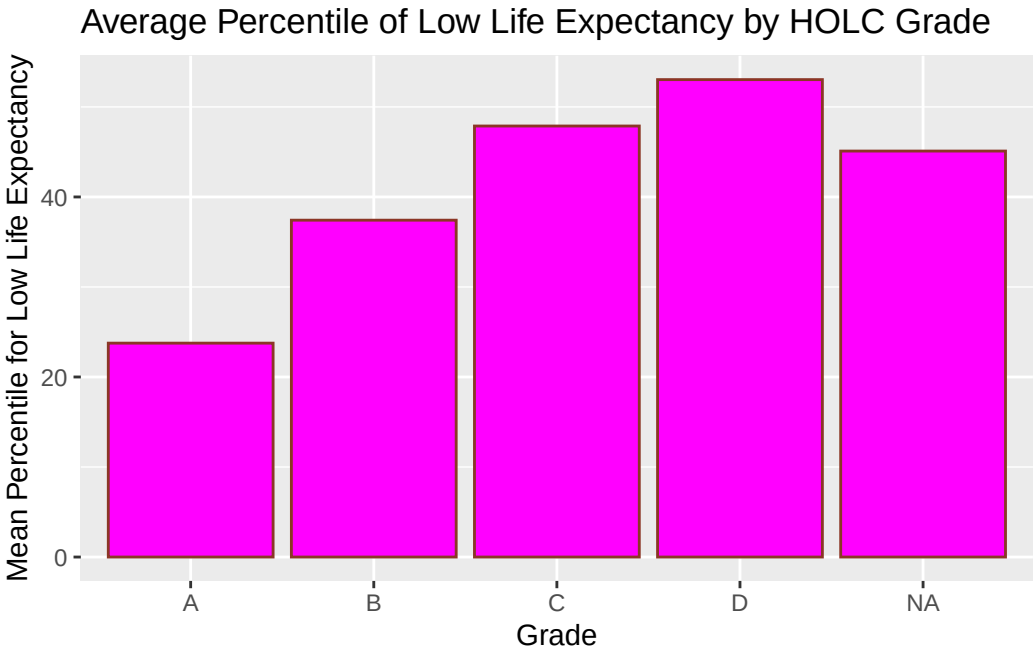
ggplot(sumbygrade, aes(x = grade, y = meanlowinc))+
    geom_bar(stat ='identity', fill = 'lightgoldenrodyellow', color = 'deeppink4') +
    labs(x = 'Grade',y = 'Proportion',
        title = 'Average Proportion Low Income in Each HOLC Grade')
```




```
ggplot(sumbygrade, aes(x = grade, y = meanP_PM25))+
    geom_bar(stat ='identity', fill = 'plum4', color = 'springgreen2') +
    labs(x = 'Grade',y = 'Percentile of PM25 Concentration',
        title = 'Average PM25 Concentration Percentile by HOLC Grade')
```




```
ggplot(sumbygrade, aes(x = grade, y = meanlowlifexpect))+
    geom_bar(stat ='identity', color = 'tomato4', fill = 'magenta1') +
    labs(x = 'Grade',y = 'Mean Percentile for Low Life Expectancy',
        title = 'Average Percentile of Low Life Expectancy by HOLC Grade')
```



Write a brief paragraph reflecting on these results

Visually, it seems that there is a relationship between the HOLC grade of a census block and % of people deemed low income, PM25, and percent of low life expectancy. The relationship is less obvious for PM25, which makes sense since air flow is less defined by neighborhood boundaries.

Part 2: Legacy of redlining in pedstrian and bike accidents

Figure summarizing the percent of observations within redlined neighborhoods within each HOLC grade

```
# Filter crash data to 2020-2025
accidentyearfilter <- crashes |>
    filter(PEDESTRIAN_ACCIDENT == 'Y' | BICYCLE_ACCIDENT == 'Y') |>
    filter(ACCIDENT_YEAR >= 2020)

unique(accidentyearfilter$ACCIDENT_YEAR)
```

```
[1] 2025 2020 2021 2024 2023 2022
```

```
# Check which columns have more NA's x/y or lat/long

sum(is.na(accidentyearfilter$POINT_X))
```

```
[1] 1634
```

```
sum(is.na(accidentyearfilter$LATITUDE))
```

```
[1] 21139
```

```
# Way more NAs in lat/lon, lets use x y instead

# Get rid of NA's in coordinates
accidentyearfilter <- accidentyearfilter |>
    filter(POINT_X != 'NA')
# Convert character columns to geometries

accidentyearfilter <- st_as_sf(accidentyearfilter,
    coords = c('POINT_X', 'POINT_Y'), crs = 4326)
```

```
# Joining accident df with HOLC in LA DF

# Check CRS's before spatially joining
if (st_crs(accidentyearfilter) == st_crs(mapineqlos)) {
    print('It is all good!')
} else {
    warning('It is not good the CRS do not match')
}

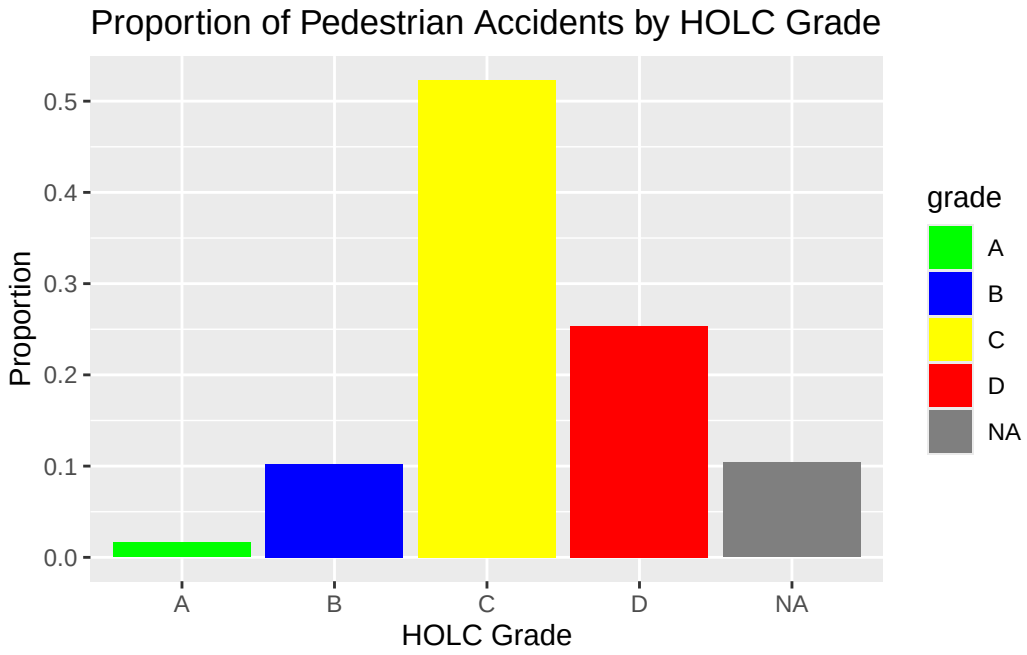

# convert CRS's and then spatial join
accidentyearfilter<- st_transform(accidentyearfilter,
    crs = st_crs(mapineqlos))

# Join accidents with HOLC in LA
accxHOLC <- st_join(mapineqlos, accidentyearfilter, left = FALSE)
```

```r
# Visualize proportion of pedestrian accidents and fatalities by HOLC Grade
proppedHOLC <- accxHOLC |>
    st_drop_geometry() |>
    filter(PEDESTRIAN_ACCIDENT == 'Y') |>
    count(grade)

proppedHOLC$propacc <- proppedHOLC$n/sum(accxHOLC$PEDESTRIAN_ACCIDENT == 'Y',
    na.rm = TRUE)

ggplot(proppedHOLC, aes(x = grade, y = propacc, fill = grade)) +
    geom_bar(stat = 'identity') +
    labs(title = 'Proportion of Pedestrian Accidents by HOLC Grade',
        x = 'HOLC Grade',y = 'Proportion') +
    scale_fill_manual(values = c('green','blue','yellow','red', 'grey'))
```




```r
# Visualize proportion of bicycle accidents by HOLC Grade

propbikeHOLC <- accxHOLC |>
    st_drop_geometry() |>
    filter(BICYCLE_ACCIDENT == 'Y') |>
    count(grade)
```

```r
propbikeHOLC$propacc <- propbikeHOLC$n/sum(accxHOLC$BICYCLE_ACCIDENT == 'Y',
    na.rm = TRUE)

ggplot(propbikeHOLC, aes(fill = grade, x = grade, y = propacc)) +
    geom_bar(stat = 'identity') +
    labs(title = 'Proportion of Bicyclist Accidents by HOLC Grade',
        x = 'HOLC Grade',y = 'Proportion') +
    scale_fill_manual(values = c('green','blue','yellow','red', 'grey'))
```

